

Testing Document

How accuracy and reliability were tested — Learning Analysis Tool

1. Testing strategy

Testing separates the two kinds of correctness this tool must have: **deterministic correctness** (parsing and scoring must be exactly right, every time) and **analytical accuracy** (the LLM's understanding judgements must match human judgement closely). Each is tested differently.

- **Unit tests** (JUnit 5) lock down parsing and scoring against known inputs.
- **Golden-sample tests** run the full pipeline on a hand-labelled session and check the ranking and weak-spot outputs.
- **Accuracy validation** (separate report) compares tool output to a human answer key.
- **Edge-case tests** cover empty chat, malformed lines, and colons inside messages.

2. Test coverage

Area	What is verified	Cases
Chat parsing	Grouping, ordering, junk rejection, colons in text	4
Scoring math	0% / 100% / mixed / partial-is-gap	4
Class weak spots	Ranking by miss count	1
End-to-end	Full pipeline on golden sample	1

3. Representative test cases

#	Scenario	Input	Expected	Result
T1	Full mastery	4 concepts all mastery	100%, 4 strengths	PASS
T2	No understanding	4 concepts all absent	0%, 4 gaps	PASS
T3	Mixed levels	mastery+understood+partial+absent	50%, 2 gaps, 2 str	PASS
T4	Partial = gap	1 concept partial	in gaps, not strengths	PASS
T5	Colon in message	'ratio is 3:1 here'	message intact	PASS
T6	Junk lines	system + blank lines	only real chat parsed	PASS
T7	Empty chat	"	empty result, no crash	PASS

4. Positive & negative scenarios

Positive: a strong student (Priya) whose chat shows correct explanations scores high with concepts in strengths. **Negative:** a confused student (Amit) whose chat shows a misconception ('photosynthesis just

makes food') scores low with concepts in gaps. The tool correctly ranks Priya above Amit — verified in the end-to-end test.

5. Reliability measures

- Deterministic mock mode lets the whole app and test suite run without an API key, so scoring/parsing regressions are caught in CI.
- LLM output is fence-stripped and parsed into typed records; malformed JSON fails loudly rather than silently.
- Prompts demand a fixed JSON schema, reducing output variance.

6. How to run the tests

```
mvn test
```

All unit, edge-case, and end-to-end tests execute against deterministic data. No API key required.